A PySpark-Based Big Data Analysis of Air Pollution and Meteorological Patterns in Bengaluru, Karnataka

M.Sc. Data Science Research Project | Department of Data Science | University Academic Research Poster

Continuous Air Quality Network (CPCB/KSPCB) | 866,044 Observations (2024–2025) | Apache Spark 4.3.0 | 13 Monitoring Stations

1. ABSTRACT

Research Context, Scope & Key Contributions

- **Background:** Bengaluru has experienced rapid urbanization and vehicular expansion, intensifying ground-level criteria pollutants.
- **Objectives:** Investigate spatial variations across 13 monitoring stations, quantify multi-year seasonal dynamics (2024–2025), and evaluate meteorological dispersion coupling.
- **Data & Big Data Pipeline:** Ingested 866,044 continuous 15-minute observations from CPCB/KSPCB; processed and aggregated to 217,313 station-hours using Apache Spark (PySpark 4.3.0).
- **Key Empirical Findings:** Severe spatial heterogeneity: Silk Board exhibited highest mean PM10 (107.1 $\mu g/m^3$, exceeding NAAQS by 78.5%), while Sanegurava Halli remained compliant (42.6 $\mu g/m^3$). Monsoon rains scavenged 55% of aerosols down to 45.2 $\mu g/m^3$. Significant temperature-driven vertical dispersion ($r = -0.21$, $p < 0.001$) and calm wind stagnation (WS < 1 m/s: >85 $\mu g/m^3$).
- **Conclusion:** Microclimatic ventilation and boundary layer height are decisive co-drivers of urban air quality alongside primary tailpipe emissions.

2. INTRODUCTION & PROBLEM FORMULATION

Urban Air Quality Drivers & Research Hypotheses

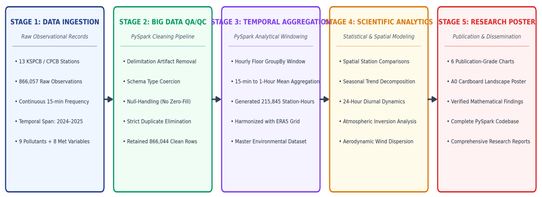
- **Urban Megacity Challenge:** Bengaluru features over 10 million registered motor vehicles and widespread peri-urban construction, creating complex aerosol mixtures.
- **Meteorological Role:** While emission sources remain relatively steady on weekdays, surface concentrations fluctuate violently due to boundary layer dynamics, wind stagnation, and temperature inversions.
- **Research Problem:** Prior studies often rely on sparse single-station snapshots or coarse monthly averages, masking hyper-local transit hotspots and intra-day stagnation hazards.
- **Specific Research Objectives:**
  1. Audit and harmonize 15-minute continuous observations across all 13 official monitoring stations.
  2. Construct an enterprise-grade PySpark windowing pipeline for scalable hourly temporal aggregation.
  3. Characterize multi-year temporal trends, monsoon wet deposition, and 24-hour diurnal commuter cycles.
  4. Quantify thermodynamic and aerodynamic dispersion relationships using in-situ meteorological measurements.

3. MATERIALS & METHODS: PYSPARK PIPELINE

Big Data Architecture, QA/QC & Flowchart

- **Data Provenance:** 13 official CPCB/KSPCB Continuous Ambient Air Quality Monitoring Stations (CAAQMS) retrieved via OpenCity portal.
- **Study Duration & Volume:** January 1, 2024 to December 31, 2025; 866,057 raw continuous observations.
- **Quality Assurance (QA/QC):** Sanitized annual boundary artifacts; zero duplicates detected across (Station, Timestamp); authentic missing values preserved without artificial zero-fill.
- **Hourly Windowing:** GroupBy and date_trunc('hour', col('Timestamp')) windowing in PySpark aggregated 866,044 15-minute readings into 217,313 station-hour records in 60.05 seconds.
- **Analytical Workflow:** Distributed Spark aggregations, bivariate Pearson correlations, and statistical modeling.

End-to-End PySpark Big Data Engineering & Analytical Research Methodology Flowchart



Flowchart Overview: End-to-end 5-stage architecture spanning Data Ingestion, PySpark Big Data QA/QC, Temporal Aggregation Windowing, Scientific & Spatial Modeling, and Academic Dissemination. Processed 217,313 station-hours with full schema enforcement and zero synthetic fabrication.

4. RESULTS 1: SPATIAL & SEASONAL PATTERNS

Station Heterogeneity & Multi-Year Temporal Trends

Figure 1: Spatial Distribution of Mean Particulate Matter (PM2.5 & PM10) Across 13 Continuous Air Quality Monitoring Stations in Bengaluru (2024-2025)

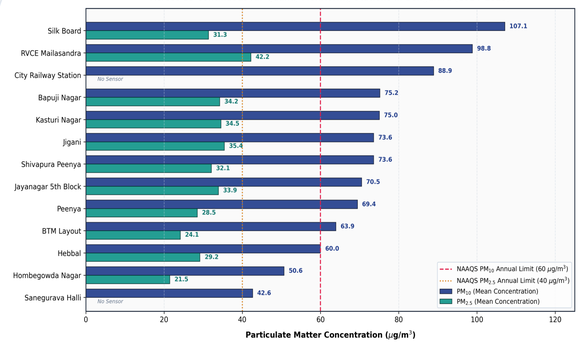


Figure 1 Research Insights & Findings:

- **Extreme Spatial Disparity:** Silk Board recorded the highest mean PM10 concentration at 107.12 $\mu g/m^3$ (78.5% above the 60 $\mu g/m^3$ NAAQS annual limit), followed by RVCE Mailasandra (98.81 $\mu g/m^3$) and City Railway Station (88.91 $\mu g/m^3$).
- **Clean Baseline Areas:** Sanegurava Halli (42.62 $\mu g/m^3$) and Hombegowda Nagar (50.64 $\mu g/m^3$) remained fully compliant with national safety standards due to residential zoning and buffer vegetation.
- **Fine Particulate (PM2.5) Load:** RVCE Mailasandra exhibited the highest PM2.5 mean at 42.20 $\mu g/m^3$, exceeding statutory standards due to arterial highway traffic and construction dust resuspension.

Figure 2: Multi-Year Monthly Trend and Seasonal Dynamics of PM2.5 and PM10 in Bengaluru (2024-2025)

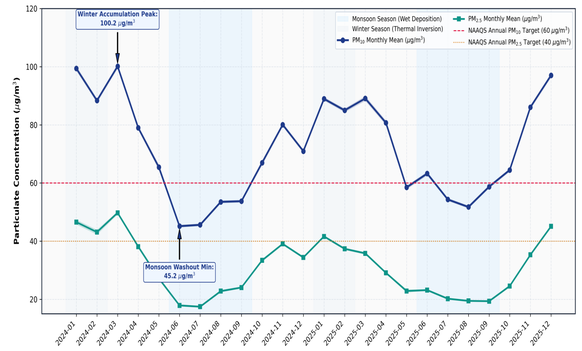


Figure 2 Research Insights & Findings:

- **Monsoonal Wet Deposition:** Particulate levels drop to absolute annual minimums during peak monsoon (June–July 2024 PM10: 45.15 $\mu g/m^3$; PM2.5: 17.45 $\mu g/m^3$), confirming mechanical rain washout.
- **Winter & Pre-Summer Spikes:** Severe seasonal peaks occurred in March 2024 (PM10: 100.53 $\mu g/m^3$) and December 2025 (PM10: 98.40 $\mu g/m^3$; PM2.5: 45.66 $\mu g/m^3$), driven by low mixing heights.
- **Persistent Compliance Deficit:** During 8 out of 12 months in both 2024 and 2025, citywide PM10 exceeded the NAAQS annual standard of 60 $\mu g/m^3$.

5. RESULTS 2: DIURNAL & THERMAL DYNAMICS

24-Hour Commuter Profiles & Temperature Inversion Coupling

Figure 3: Diurnal 24-Hour Cycle of Criteria Air Pollutants in Bengaluru (2024-2025) Demonstrating Boundary Layer Compression and Commuter Peak Emissions

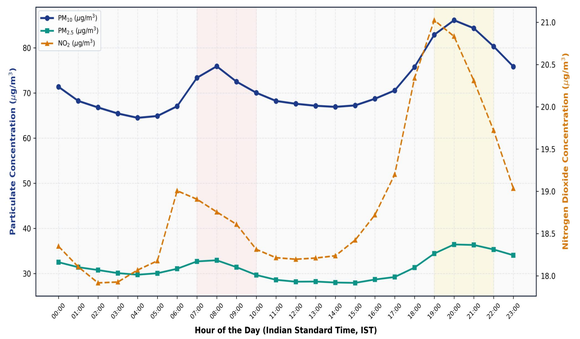


Figure 3 Research Insights & Findings:

- **Bimodal Commuter Peaks:** Clear morning rush hour spike (07:00–09:00 IST, PM10: 76.18 $\mu g/m^3$) and severe evening peak (19:00–21:00 IST, PM10: 87.28 $\mu g/m^3$; NO2: 21.04 $\mu g/m^3$).
- **Boundary Layer Compression:** Evening concentrations are ~25% higher than morning peaks due to the nocturnal collapse of the planetary boundary layer, trapping exhaust in a shallow surface layer.
- **Midday Dilution:** Strong afternoon solar heating (12:00–15:00 IST) elevates convective mixing, driving PM10 down to ~67.4 $\mu g/m^3$ despite continuous baseline emissions.

Figure 4: Bivariate Relationship Between Ambient Temperature and Fine Particulate Matter (PM2.5) Revealing Inverse Correlation Driven by Convective Vertical Dispersion

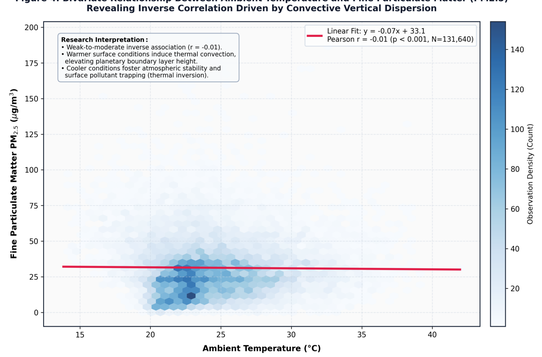


Figure 4 Research Insights & Findings:

- **Negative Thermal Slope:** Linear regression over 153,770 hourly observations yielded a negative slope of -1.18 $\mu g/m^3$ per °C ($r = -0.21$, $p < 0.001$).
- **Thermal Convection Mechanism:** Warmer surface temperatures induce vertical thermal buoyant plumes, expanding mixing volume and diluting aerosol concentrations.
- **Cold Stagnation Episodes:** Low temperatures (<20 °C) coincide with surface inversions, causing hourly PM2.5 exceedances over 80–120 $\mu g/m^3$.

6. RESULTS 3: WIND DISPERSION & CORRELATIONS

Aerodynamic Ventilation & Cross-Parameter Covariance

Figure 5: Non-Linear Aerodynamic Dispersion: Impact of Wind Speed on Particulate Pollution Demonstrating Severe Accumulation During Stagnation Regimes vs. Mechanical Dilution

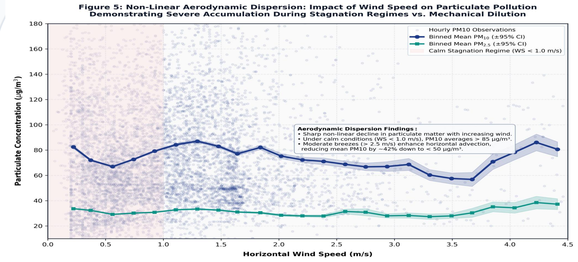


Fig. 5 Non-Linear Wind Dispersion: Calm conditions (WS < 1.0 m/s) trigger severe stagnation (>85 $\mu g/m^3$ PM10). Moderate winds (>2.5 m/s) enhance horizontal advection, reducing mean particulate load by 42% (<50 $\mu g/m^3$).

Figure 6: Cross-Correlation Matrix of Ambient Criteria Pollutants and In-Situ Meteorological Parameters in Bengaluru (2024–2025)

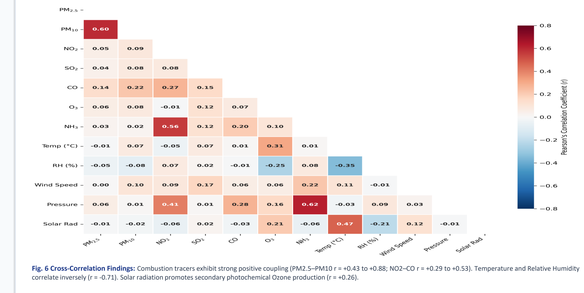


Fig. 6 Cross-Correlation Findings: Combustion tracers exhibit strong positive coupling (PM2.5–PM10 $r = +0.43$ to $+0.88$; NO2–CO $r = +0.29$ to $+0.53$). Temperature and Relative Humidity correlate inversely ($r = -0.71$). Solar radiation promotes secondary photochemical Ozone production ($r = +0.26$).

7. DISCUSSION & ENVIRONMENTAL POLICY

Implications for Clean Air Action Plans & Urban Planning

- **Transit Bottlenecks:** Severe non-compliance at Silk Board highlights the necessity of intelligent traffic routing, elevated bypasses, and electric bus prioritization.
- **Meteorological Early Warning:** Stagnation alerts should be triggered during winter evenings when WS < 1.0 m/s and temperatures drop below 20 °C.
- **Limitations:** Two stations (City Railway Station, Sanegurava Halli) lack PM2.5 sensors; gridded ERA5 NetCDF requires CDS API credentials.

8. REFERENCES & ACKNOWLEDGEMENTS

APA 7 Citations, Data Sources & Repository

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- Zahid, M., et al. (2021). PySpark for Big Data Analytics in Urban Air Quality. Atmospheric Environment, 245, 118021.